

# Etienne Marion

📍 Office 725 C1, 45 rue des Saints-Pères, 75006 Paris    ✉ [etienne.marion@math.cnrs.fr](mailto:etienne.marion@math.cnrs.fr)    🌐 [etiennec30.github.io](https://etiennec30.github.io)

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## Summary

I am a PhD student in Mathematics at MAP5, Université Paris-Cité. I am mainly interested in probability theory applied to population genetics and formalization of Mathematics using the proof assistant Lean.

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## Education

- 2026 – present    **MAP5, Université Paris-Cité**, PhD in probability applied to population genetics – Paris, France  
Supervised by Félix Foutel-Rodier
- 2025 – 2026    **University of Vienna & Université Paris-Cité**, probability applied to population genetics  
Research internships
- 2023 – 2025    **École Normale Supérieure de Lyon**, MS in probability and statistics – Lyon, France
- 2022 – 2023    **École Normale Supérieure de Lyon**, BS in mathematics – Lyon, France
- 2020 – 2022    **Lycée Georges Clemenceau**, Classe préparatoire in mathematics and physics – Reims, France

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## Research experience

- Feb 2026 – July 2026    **Research intern**, MAP5, Université Paris-Cité – Paris, France  
Supervised by Félix Foutel-Rodier: *Impact of demographic fluctuations on structured populations genealogies.*
- Sept 2025 – Jan 2026    **Research intern**, University of Vienna – Vienna, Austria  
Supervised by Emmanuel Schertzer: *Diffusive neutral fractions inside a genetic population.*
- Mar 2025 – July 2025    **Research intern**, École Normale Supérieure de Lyon – Lyon, France  
Supervised by Jean-Christophe Mourrat: *Study of functions learnable by neural networks using mean-fields approximation.*
- May 2024 – July 2024    **Research intern**, Institut de recherche mathématique de Rennes – Rennes, France  
Supervised by Sébastien Gouëzel: *Formalization of the Ionescu-Tulcea theorem and construction of infinite products of probability measures with Lean.*
- May 2023 – June 2023    **Research intern**, Laboratoire Jean Kuntzmann – Grenoble, France  
Supervised by Sana Louhichi: *Tight sequences of random variables: definition, examples and application.*

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## Publications

- 2026    **A Formalization of the Ionescu-Tulcea Theorem in Mathlib**  
Etienne Marion  
[link.springer.com/journal/10817](https://link.springer.com/journal/10817) (Journal of Automated Reasoning)

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## Preprints

2025+ **Formalization of Brownian motion in Lean**  
Rémy Degenne, David Ledvinka, Etienne Marion, Peter Pfaffelhuber  
[arxiv.org/abs/2511.20118](https://arxiv.org/abs/2511.20118)

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## Talks

Jan 2026 **A Formalization of the Ionescu-Tulcea Theorem in Mathlib**  
Lean Together 2026  
[www.youtube.com/watch?v=\\_TrTKnWU-to](https://www.youtube.com/watch?v=_TrTKnWU-to)

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## Participation to conferences

**Sept 2026:** I was present at the Besançon meeting on Probability, Ecology & Evolution in Besançon.

**Apr 2026:** I was present at the 3rd annual meeting of the GdR Branchement in Avignon.

**Sept 2025:** I was present at the workshop Ancestral lines in population models with interaction at BOKU University, in Vienne.

**June 2025:** I was present at the summer school Local Limits of Random Graphs at Institut de Mathématique d'Orsay.

**Mar 2024:** I was present at the conference Lean for the curious mathematician at CIRM.

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## Other activities

I am part of the Mathlib reviewer team. I am assessing the mathematical relevance and the quality of the code that contributors wish to add to the library, especially in the probability section.

**Sept - Dec 2023:** With three fellow students we gave an introductory course to category theory, shaped as a weekly seminar.